Brick by Brick: Using Collaborative Robotics to Combat Social Isolation in Public Spaces

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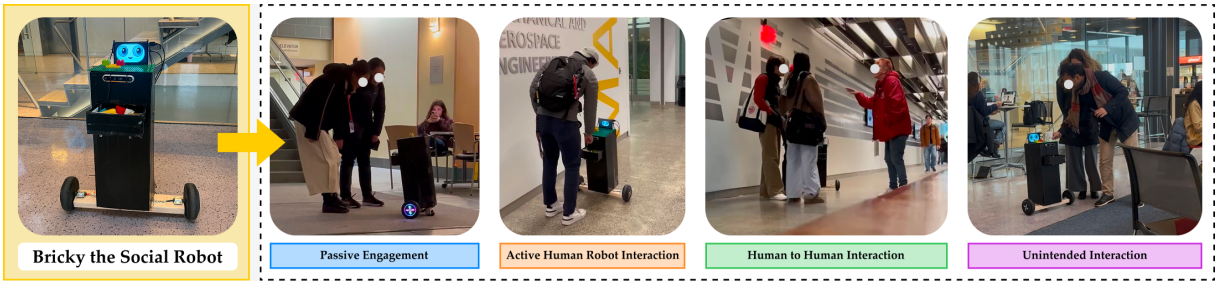


Figure 1: Interaction patterns discovered during field testing

Abstract

Social isolation in shared spaces threatens community cohesion and well-being. This paper presents a social robot designed to spark human-to-human interactions. Inspired by public art projects, the robot invites individuals to collaborate on a shared LEGO structure by using expressive eye tracking, autonomous turning, and servo-actuated drawer movement. Field deployments in Cornell University spaces showed the robot effectively acted as a social catalyst: diverse participants contributed to a shared structure, and strangers initiated conversations about the robot. This work offers a functional prototype and insights on robots as mediators of human connection and promotes ideas of empowering collaboration.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design**.

Keywords

Human-Robot Interaction, Social Robotics, Collaborative Art

ACM Reference Format:
Joyce Yang, Phillip Johnson, Nyra Graham, and Karen Shamir. 2026. Brick by Brick: Using Collaborative Robotics to Combat Social Isolation in Public Spaces. In *Companion Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI Companion '26), March 16–19, 2026, Edinburgh, Scotland, UK*. ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3776734.3794595

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HRI Companion '26, Edinburgh, Scotland, UK
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ACM ISBN 979-8-4007-2321-6/2026/03
https://doi.org/10.1145/3776734.3794595

1 Introduction

Walk into any university shared work space or building lobby, and a pattern emerges: people sitting feet apart, yet entirely disconnected, using screens as shields against social interaction. This widespread retreat from spontaneous connection represents a growing social crisis. As recent reporting shows [4], American adults now spend 30% less time in face-to-face socializing than two decades ago, with teenagers declining 45%. Research on "relational diversity" shows that interactions with varied relationship categories, including acquaintances and strangers, significantly boost well-being [1]. Social robots deployed in museums, malls, and airports have shown promise engaging people in natural contexts [2, 3], while collaborative public art such as physical community murals and Reddit's digital canvas "r/place" demonstrates that shared creation fosters social cohesion [5, 6]. However, translating collaborative art principles into mobile, dynamic shared spaces remains largely unexplored. This raises a critical design challenge: How can we gently disrupt social inertia and transform shared spaces into places of spontaneous connection? In this paper, we propose and evaluate a **social robot that roams campus spaces and invites passersby to contribute LEGO bricks to a shared structure.** Drawing on principles from social robotics and collaborative art, our system acts as a "social mediator," creating "triangulated" interactions where the robot and shared building activity provide a focal point facilitating human connection. Through simple play and shared creation, we aim to spark micro-interactions that rebuild the casual social fabric missing from our public spaces.

2 Problem Statement

The challenge we address is how to create opportunities for human-to-human interaction in shared spaces where such interactions have become socially uncomfortable or rare. To address this challenge, we established the following design goals for our robot:

(1) **Low Barrier to Participation:** The interaction must be accessible to all people.
(2) **Curiosity-Driving:** The robot should naturally attract attention and invite questions through its design.
(3) **Collaborative, Not Individual:** The robot must create a shared experience and artifact that multiple people contribute to over time.
(4) **Mobile and Proactive:** The robot must actively initiate interaction with people.
(5) **Autonomous:** The robot must be fully capable of making decisions and powering movement without intervention from a researcher.

3 Design & Interaction

3.1 Low-Fidelity Prototypes

In the initial stages of development, we prioritized testing a critical question: Can we engage passersby in a public space with a static robot featuring only actuated movement, a LEGO base plate, and LEGO bricks? The basic structure of our prototypes consisted of a table-like form where a person could hide underneath and operate a sliding platform to present LEGO bricks to participants. The top surface featured a LEGO base plate where pieces could be placed. We iterated over design choices about drawer location and gestures as well as the inclusion of eyes that were either static or followed pedestrians as they passed by.



Figure 2: Low-fidelity prototypes initial sketches and in field testing

3.2 High-Fidelity Prototype

These low-fidelity tests yielded several critical insights that informed our high-fidelity design. First, the front-facing drawer placement was essential for communicating interaction opportunities. Second, visual face elements were necessary to establish the robot as a social agent worthy of engagement rather than passive furniture. Third, while the static prototype successfully garnered curiosity, the completion rate of full interactions, such as taking a LEGO brick and placing it on the base plate, remained low. This suggested that mobility would be crucial for our high-fidelity design: a mobile robot could signal invitation more clearly than a stationary one. Using these insights, we developed the following **interaction flow** as seen in Figure 3.

(1) **Approach:** The robot identifies individuals in a public space and turns toward them.
(2) **Engage:** As the robot turns, its animated eyes track the person's position. The drawer slides open, revealing the LEGO bricks inside and gestures for the person to take a piece.
(3) **Participate:** The person takes a LEGO brick from the drawer and places it on the base plate.
(4) **Attract:** As the LEGO structure on the base plate grows, new arrivals see the partially-built structure and approach the robot.
(5) **Connect:** The robot becomes a natural conversation topic. Strangers ask each other "What is this?" or "Did you add a piece?" or comment on the emerging form.



Figure 3: Interaction flow storyboard sketch

3.3 Hardware

After deciding on our high-fidelity design, we chose hardware that allowed for high customizability and enabled our ambitious goals. To control motion, an ODrive motor controller was fitted to wheels of a Hover-1 Blast retail hoverboard. These wheels were controlled using ODrive Robotics, Inc.'s Python library under the MIT license through a Raspberry Pi 5 running Ubuntu 24.03. Also connected to the Raspberry Pi was an Intel D455 Realsense stereo and depth camera that was used for person detection and distance measurements.

Another interaction element of the robot was the mobile drawer that was controlled through a servo and a linear actuator. These allowed for customizable "gestures" through efficient drawer movement. The servo was controlled through an Arduino Uno that was connected over serial connection to the Raspberry Pi.

The final piece of hardware in the autonomy stack was a 7" Raspberry Pi screen that displayed the eyes of the robot and followed people that were detected through the Realsense camera. The Raspberry Pi and its peripherals were powered through a portable power bank, and the ODrive was powered through a Lithium-ion battery pack.

3.4 Software

With the heterogeneity of the system, it was imperative that the software behind it was maintainable, efficient, and reproducible. This section will detail the decisions and algorithms that allowed for autonomy and safety.

3.4.1 Robotics Operation System (ROS2). Due to the complex interactions between hardware and software, ROS2 middleware was used to ensure high-throughput communication between each component. The overview of this system can be seen in Figure 4. The circles in this figure represent nodes in the network, and the arrows represent the direction of data flow.

3.4.2 Person Detection. TensorFlow Lite's mobilenet person detection machine learning library [7] was used under the Apache 2.0 License to produce bounding boxes of detected people. Then, using the depth camera, the closest person was detected and that data was published to the corresponding nodes. The robot's point of view from the person detection output can be seen in Figure 5.

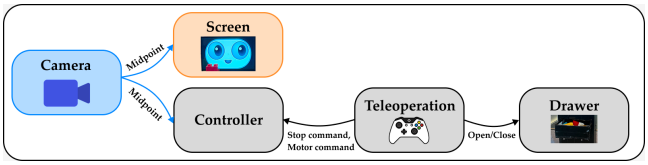


Figure 4: ROS2 network overview

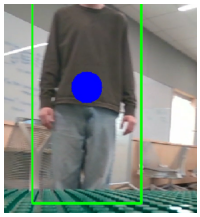


Figure 5: Bounding box and midpoint output of model

3.4.3 Autonomous Control. Once the closest person was detected, the midpoint was fed into a Linear-Quadratic-Regulator (LQR) heading controller to control the robot to face them. LQR uses the system's dynamics, as defined by the researcher, to determine the globally optimal compensator gains based on a selection of cost matrices. To do this, the algebraic Ricatti differential equation (1) is solved for the P matrix based on cost matrices, Q and R, which determine the penalty for the difference between the desired state and the overall control effort. This P matrix is used to determine the gain matrix K (2) which is fed into the compensator for control output u (3). Once the heading was controlled, a simple, low-speed proportional controller was used to drive forward to the person and stop at a predefined safety distance.

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0 \tag{1}$$

$$K = R^{-1}B^\top P \tag{2}$$

$$u = u_0 - Kx \tag{3}$$

3.4.4 Manual Control (Teleoperation). For teleoperation, an Xbox Series X Bluetooth controller was connected to the Raspberry Pi. The drawers and wheels could be manually commanded through this controller, and it was used in the cases of a deadlock (i.e. the robot is facing a wall), stop command, or drawer command. During field testing, controller use was minimal to ensure that interactions were driven through robot autonomy and allowed the researchers to blend into the environment and promote natural reactions.

3.4.5 Safety. When building an autonomous robot, safety features must be put in place before any control can occur. This robot has a tiered safety system to ensure that it will not collide with a human under any circumstances. The safety structure is outlined as follows:

(1) The robot only moves once a person is detected in the frame.
(2) The robot will always stop when a person is detected to be within stopping distance.
(3) A human teleoperator can stop the robot at any time through joystick control.

Throughout field testing, there were zero collisions due to the structure of this safety controller and robust handling of any error states.

4 Results & Discussion

4.1 Results

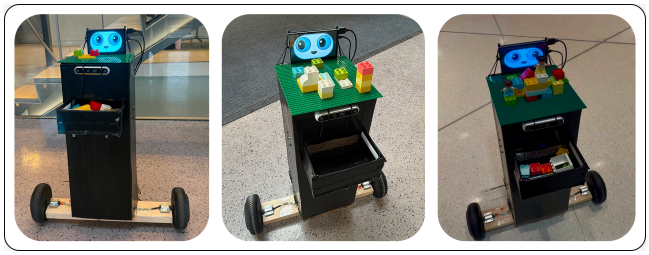


Figure 6: From left to right: the robot pre-deployment at Gates Hall, after a 15-minute session at Gates Hall, and after a 15-minute session at Statler Lobby.

4.1.1 Field Testing Methodology. We deployed our robot in three campus locations: Duffield Atrium (engineering building workspace), Statler Lobby (business school building), and Gates Hall (computer science building café). Sessions lasted 15-30 minutes during moderately populated weekday afternoons and evenings. We collected data through video recordings from multiple angles and detailed field notes documenting interaction patterns. All observations were conducted in accordance with institutional guidelines for public space research.

4.1.2 Observed Interaction Patterns. In Figure 1, we identified four distinct categories of engagement that emerged across our deployments ranging from passive observation to active collaborative participation.

(1) **Passive Engagement:** Individuals paused to observe or photograph the robot. Passive observers often formed small clusters creating opportunities for brief exchanges.
(2) **Active Human-Robot Interaction:** Beyond taking LEGO bricks and placing them on the board, participants attempted verbal communication with the robot, inspected mechanical components, and tested responsiveness by waving or touching the drawer. These exploratory behaviors indicated the robot invited safe, playful investigation.
(3) **Human-Human Interaction:** The robot catalyzed conversations between strangers through shared observation commentary ("That's so cute!"), information sharing (explaining the robot's purpose to newcomers), and participation encouragement (inviting others to add a brick).
(4) **Unintended Interactions:** During field testing, we observed people using the robot as a coffee stand, nudging it with their foot, and even knocking it over.

4.2 Discussion

Our field deployments revealed critical insights about designing robots to facilitate human connection in public spaces.

Table 1: Field Deployment Statistics Across Three Locations

Metric	Duffield Atrium	Statler Lobby	Gates Hall
Duration	15 min	15 min	15 min
People Passed By	~50	~93	~23
Passive	~44	~23	~15
Active	5	8	5
Human-to-Human	2	4	0

4.2.1 Design Insights. A critical design insight was the notion of "allowances" in human–robot interaction, meaning features that resemble everyday objects and function as expected, familiar interaction elements. One example of this was a participant placing their coffee cup on the robot as it resembled a table located in the café. This design insight was what drove our decision-making behind the use of LEGO bricks. Their immediate recognizability transcended demographic boundaries, and the physical act of snapping bricks provided satisfying tactile feedback that digital interactions cannot replicate. Importantly, LEGOs carry no performance anxiety: there's no "wrong" way to place a brick, making participation psychologically safe. Additionally, mobility proved helpful but not critical. While we hypothesized it would be essential for initiating interactions, even stationary deployments with the actuated drawer and eye tracking successfully attracted participants.

4.2.2 Toward Future Designs. In general, roboticists have a tendency to focus on the mathematically optimal approach toward accomplishing a task. In our field studies, we found that these more optimal approaches could be a hindrance toward natural interactions. A primary example of this was the design of the control system. In the lens of optimality, it would make the most sense to design the system so that the total control cost is minimized. However, we found that the more expensive approach of turning the robot toward participants before approaching closer simulated how humans interact. Additionally, prior research reinforces the idea that animating motion makes a robot more approachable [8], and our "facing" motion was inspired by this principle. When engineering robots that exist around humans, it is imperative to consider how the motion is perceived, and we found that this early, non-intrusive movement established trust and promoted approachability.

5 Conclusions

5.0.1 Social Insights. From our field deployments, there were several key, unexpected insights. First, traffic volume does not guarantee increased interaction. We deployed the robot in variously trafficked spaces. Surprisingly, in crowded, rushed environments, people often hurried past the robot with brief glances but minimal engagement. Conversely, in calmer spaces where individuals were already lingering, we observed deeper engagement. In addition, setting and demographics significantly influence interaction patterns. In the Computer Science building, while passive engagement remained frequent, active participation was substantially lower and most passersby showed muted reactions. In contrast, in the Business school building, reactions were far more pronounced, with visible expressions of surprise, happiness, and wonder. Participants were engaged in significantly more direct human-robot interactions and, critically, more human-to-human conversations about the robot. Lastly, we hypothesize that goal-oriented building may foster stronger connections than curiosity alone. Future iterations might explicitly prompt collective building goals—such as "help build a rainbow" or "create the tallest tower"—to catalyze deeper stranger collaboration.

5.0.2 Empowering Society through combating social isolation: Social isolation manifests not only as a lack of close relationships but as the chronic absence of casual human acknowledgment in daily life. While our robot may not create deep relationships, it sparks micro-interactions, as evidenced by our field deployments. Through repeated presence, such robots could gradually shift social norms from stranger avoidance toward casual engagement. Our robot has the potential to transform shared spaces from collections of isolated individuals into sites of potential connection, making libraries, lobbies, and cafés feel more welcoming and less alienating. While our testing focused on a university campus, the core approach applies to many contexts, including hospital waiting rooms where patients sit anxiously alone, senior centers where older adults face chronic loneliness, and transit hubs where commuters experience daily anonymity. Future work will explore how the design can be adapted across these diverse settings to restore the casual social fabric that makes shared spaces genuinely communal.

Acknowledgments

This research was conducted as a course project. The authors thank Dr. Malte Jung and Michael Sack for their guidance. This work was supported by the National Science Foundation (CAREER 1942085) and the Office of Naval Research (N00014-19-1-2299). Any opinions, findings, conclusions, or recommendations expressed are those of the authors and do not necessarily reflect the views of the funding agencies.

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Received 2025-12-08; accepted 2026-01-12